

Senior Software Engineer for 10+ years building award-winning systems across fintech, healthcare, and national security.

Professional Experience

Founder & Principal Engineer ▪ Spirit & Hammer

October 2025 - Present

Technologies: *Next.js, React, TypeScript, Python, FastAPI, Vercel AI SDK, OpenRouter, Claude, Three.js, Cloudflare, Swift, Playwright*

- Shipped full-stack AI circadian-health platform that turns any US zip code into real-time daily timing guidance, backed by a AI chat and 50K-LOC FastAPI engine, dynamic render Vercel AI UI components, and an MCP server exposing 14 tools to LLMs.
- Engineered offline-first, agent-editable apps on Cloudflare Workers with local-first sync, conflict resolution, and custom HMAC magic-link auth, all controllable through SSE-streaming AI chat with full undo.
- Architected end-to-end AI video pipelines that convert audio and transcripts into fully animated 1080p videos, replacing manual editing with automated scene segmentation, illustration, cinematography, and multilingual dubbing. Built a YouTube intelligence platform with a multi-provider LLM abstraction and enrichment to improve the player experience.

Principal Software Engineer ▪ Capital One

Jan 2021 to Dec 2021, Aug 2024 to Oct 2025

Technologies: *Java, JavaScript, TypeScript, SolidJS, React, Angular, Spring Boot, AWS, Docker, Git, REST APIs, JUnit, AI Tools, Co-Pilot*

- Generated millions in revenue by delivering Auto Navigator Trade-In in two months with an understaffed team—led cross-functional alignment and architected scalable document upload.
- Built full-stack features (Trade-In, Auto Loan Rates, Search) serving millions of users with React, SolidJS, Angular, and Spring Boot microservices on AWS.
- Cut code-review cycles 40% and eliminated 100K API failures by leading cybersecurity remediation as Duty Champion with automated security testing.
- Enabled zero-defect deployments for 3M+ daily users by modernizing a legacy Angular codebase with automated testing and disciplined releases.
- Won a company-wide hackathon for UX design and frontend implementation.

Senior Software Engineer ▪ Publicis Sapient

April 2022 to May 2023

Technologies: *Angular, TypeScript, Java, Spring Boot, ArcGIS, AWS, PostgreSQL, Docker*

- Increased healthcare access 20% by building an enterprise Angular mapping app on a Java/Spring Boot backend, improving provider discovery for underserved communities.
- Scaled the platform to enterprise workloads by implementing OO design patterns and a microservices architecture for maintainable distributed systems.
- Reduced defect rates 33% by introducing automated test coverage and structured peer-review processes.
- Improved engagement and accessibility for diverse healthcare users by building multi-device responsive UIs with Angular Material.

Senior Software Engineer ▪ Maxar Technologies

Dec 2018 to Apr 2022

Technologies: *Angular, TypeScript, Java, Spring Boot, AWS, Kubernetes, REST APIs, Junit*

- Supported national decision-makers by engineering distributed Angular apps for geospatial mapping and probability simulation.
- Enabled rapid analysis of high-volume data by building real-time visualization with parallel processing in modern JS frameworks.
- Drove adoption of complex engineering solutions by partnering with technical and business stakeholders to validate requirements and deliver technical demos.

Skills & Certifications

Certifications: Claude Certified Architect (June 2026), John Hopkins Agentic AI, AWS Certified Solutions Architect

AI & ML: Agentic AI, RAG Pipelines, ML, LLM, Claude, Claude Code, Claude Certified Architect (June 2026), OpenRouter, AI SDK

Languages: Java, Kotlin, Typescript, JavaScript, Python, Node.js

Frontend: React, Angular, SolidJS, RxJS & NgRx, geospatial mapping (ArcGIS, Cesium.js, Leaflet.js, Google Maps API)

Backend & Data: Spring Boot, Flask, REST, Microservices, PostgreSQL, MongoDB, NoSQL, distributed systems, parallel computing **Cloud,**

Devops & Testing: AWS (Certified Solutions Architect), Docker, Kubernetes, CI/CD, Git, Junit/Jest/Selenium, TDD/BDD

Education

Bachelors of Science, Electrical Engineering ▪ George Mason University